United States Patent	12416144
Kind Code	B2
Date of Patent	September 16, 2025
Inventor(s)	Fang; Nathan et al.

Bidet washing apparatus with deodorizing feature

Abstract

A bidet washing apparatus includes a control unit connected to one or more water inlets configured to supply water to the control unit, the control unit having one or more control unit switches configured to operate one or more control unit valves for controlling water flow from the one or more water inlets and/or from one or more control unit outlets; a deodorizer housing unit, wherein the housing unit includes a deodorizer container, and a deodorizer pump; a deodorizer tube connected to the deodorizer pump; a deodorizer spray or drop outlet connected to the deodorizer tube; and a nozzle assembly including at least one washing nozzle connected to at least one of the one or more control unit outlets with one or more control unit outlet to nozzle assembly water tubes.

Inventors: **Fang; Nathan (San Diego, CA), Gupta; Anjum (San Diego, CA)**

Applicant: **2GO PRODUCTS, LLC** (San Diego, CA)

Family ID: **1000008821940**

Appl. No.: **17/980320**

Filed: **November 03, 2022**

Prior Publication Data

Document Identifier	Publication Date
US 20240151018 A1	May. 09, 2024

Publication Classification

Int. Cl.: **E03D9/00** (20060101); **E03D9/08** (20060101)

U.S. Cl.:

CPC **E03D9/005** (20130101); **E03D9/08** (20130101);

Field of Classification Search

CPC: E03D (9/005); E03D (9/08)

USPC: 4/420.4-; 4/420

References Cited

U.S. PATENT DOCUMENTS

Patent No.	Issued Date	Patentee Name	U.S. Cl.	CPC
7076812	12/2005	Haberkorn	4/420.4	E03D 9/08
2013/0145537	12/2012	Al-Jafar	4/420.5	A61G 5/125
2015/0184368	12/2014	Plate	4/420.4	E03D 9/005
2024/0151018	12/2023	Fang	N/A	E03D 9/005

Primary Examiner: Baker; Lori L

Background/Summary

FIELD OF THE INVENTION

(1) The disclosure generally relates to a bidet washing apparatus, and more particularly to a bidet washing apparatus having a deodorizing feature that can store and dispense a deodorizer from a bidet washing system.

BACKGROUND OF THE INVENTION

(2) A bidet apparatus for washing and cleaning body parts were initially developed in the form of a bidet that provided a single spray of water and was permanently built into the toilet bowl. However, such bidets were expensive, and a new generation of bidets was developed that was attachable to the toilet and included a plurality of nozzles for multiple water sprays. Such bidets can be attached to the seat of an existing toilet bowl for washing the private parts of a person.

(3) Various bidet designs have addressed some of the desired effects, such as washing, washing with temperature-regulated water, and drying, However, existing bidets fail to address all concerns related to the designs and functions in the general field of bidets. In addition, many users are embarrassed by odors leaving the toilet and escaping into the air. This is particularly important in bidets used, for example, in a public place or in a busy household.

(4) Currently, there aren't any bidets that include a deodorizing feature. A deodorizing feature would prevent odors from escaping into the air and should be conveniently integrated into a bidet thereby eliminating the need to install a second device and allowing the user not having to remember to spray before sitting down.

SUMMARY OF THE INVENTION

(5) The disclosed embodiments are directed to solving one or more of the problems presented in the prior art, described above, as well as providing additional features that will become readily apparent by reference to the following detailed description when taken in conjunction with the accompanying drawings.

(6) In an embodiment, the disclosure provides a bidet washing apparatus, which includes a control unit fluidically connected to one or more water inlets configured to supply water to the control unit, the control unit including one or more control unit switches configured to operate one or more control unit valves for controlling water flow from the one or more water inlets and/or from one or

more control unit outlets; a deodorizer housing unit, wherein the housing unit includes a deodorizer container, and a deodorizer pump; a deodorizer tube fluidically connected to the deodorizer pump; a deodorizer spray or drop outlet fluidically connected to the deodorizer tube; and a nozzle assembly including at least one washing nozzle fluidically connected to at least one of the one or more control unit outlets with one or more control unit outlet to nozzle assembly water tubes.
(7) In one aspect, the disclosure provides a bidet washing apparatus wherein the deodorizer housing unit, the deodorizer container, and the deodorizer pump are attachable to the control unit.
(8) In another aspect, the disclosure provides a bidet washing apparatus wherein the deodorizer pump includes a deodorizer actuator.
(9) In another aspect, the disclosure provides a bidet washing apparatus wherein the deodorizer pump is a bottle pump.
(10) In another aspect, the disclosure provides a bidet washing apparatus wherein the deodorizer pump includes a pump chamber, a spring, a ball, a piston, a pump tube inlet, and a pump tube outlet.
(11) In another aspect, the disclosure provides a bidet washing apparatus wherein the deodorizer spray or drop outlet is reversibly attachable to the nozzle assembly.
(12) In another embodiment, the disclosure provides a bidet washing apparatus, which includes a control unit fluidically connected to one or more water inlets configured to supply water to the control unit, the control unit including one or more control unit switches configured to operate one or more control unit valves for controlling water flow from the one or more water inlets and/or from one or more control unit outlets; a deodorizer container and a deodorizer pump, wherein the deodorizer container and the deodorizer pump are located within the control unit; a deodorizer tube fluidically connected to the deodorizer pump; a deodorizer spray or drop outlet fluidically connected to the deodorizer tube; and a nozzle assembly including at least one washing nozzle fluidically connected to at least one of the one or more control unit outlets with one or more control unit outlet to nozzle assembly water tubes.
(13) In one aspect, the disclosure provides a bidet washing apparatus wherein the deodorizer pump includes a deodorizer actuator.
(14) In another aspect, the disclosure provides a bidet washing apparatus wherein the deodorizer pump is a bottle pump.
(15) In another aspect, the disclosure provides a bidet washing apparatus wherein the deodorizer pump includes a pump chamber, a spring, a ball, a piston, a pump tube inlet, and a pump tube outlet.
(16) In another aspect, the disclosure provides a bidet washing apparatus wherein the deodorizer spray or drop outlet is reversibly attachable to the nozzle assembly.
(17) In another embodiment, the disclosure provides a bidet washing apparatus, which includes a control unit fluidically connected to one or more water inlets configured to supply water to the control unit, the control unit including one or more control unit switches configured to operate one or more control unit valves for controlling water flow from the one or more water inlets and/or from one or more control unit outlets; a deodorizer housing unit, wherein the housing unit includes a deodorizer container, and a deodorizer pump, and wherein the deodorizer housing unit is located distal to the control unit; a deodorizer tube fluidically connected to the deodorizer pump; a deodorizer spray or drop outlet fluidically connected to the deodorizer tube; and a nozzle assembly including at least one washing nozzle fluidically connected to at least one of the one or more control unit outlets with one or more control unit outlet to nozzle assembly water tubes.
(18) In one aspect, the disclosure provides a bidet washing apparatus wherein the deodorizer pump includes a deodorizer actuator.
(19) In another aspect, the disclosure provides a bidet washing apparatus wherein the deodorizer pump is a bottle pump.
(20) In another aspect, the disclosure provides a bidet washing apparatus wherein the deodorizer

pump includes a pump chamber, a spring, a ball, a piston, a pump tube inlet, and a pump tube outlet.
(21) In another aspect, the disclosure provides a bidet washing apparatus wherein the deodorizer spray or drop outlet is reversibly attachable to the nozzle assembly.
(22) In another embodiment, the disclosure provides a bidet washing apparatus, which includes: a control unit fluidically connected to one or more water inlets configured to supply water to the control unit, the control unit including one or more control unit switches configured to operate one or more control unit valves for controlling water flow from the one or more water inlets and/or from one or more control unit outlets; a deodorizer housing unit, wherein the housing unit includes a deodorizer aerosol container; a deodorizer tube fluidically connected to the deodorizer aerosol container; a deodorizer spray or drop outlet fluidically connected to the deodorizer tube; and a nozzle assembly including at least one washing nozzle fluidically connected to at least one of the one or more control unit outlets with one or more control unit outlet to nozzle assembly water tubes.
(23) In one aspect, the disclosure provides a bidet washing apparatus, wherein the deodorizer housing unit and the deodorizer aerosol container are attachable to the control unit.
(24) In another aspect, the disclosure provides a bidet washing apparatus, wherein the deodorizer spray or drop outlet is reversibly attachable to the nozzle assembly.
(25) In another aspect, the disclosure provides a bidet washing apparatus, wherein the deodorizer housing unit and deodorizer aerosol container are located within the control unit, or are located distal to the control unit.
(26) Further features and advantages of the disclosure, as well as the structure and operation of various embodiments of the disclosure, are described in detail below concerning the accompanying drawings.

Description

BRIEF DESCRIPTION OF THE DRAWINGS
(1) The disclosure, by one or more various embodiments, is described in detail concerning the following figures, The drawings are provided for purposes of illustration only and merely depict exemplary embodiments of the disclosure. These drawings are provided to facilitate the reader's understanding of the disclosure and should not be considered limiting the breadth, scope, or applicability of the disclosure. It should be noted that for clarity and ease of illustration these drawings are not necessarily made to scale.
(2) FIG. **1** illustrates an embodiment of a perspective view of an exemplary bidet washing apparatus installed on an existing toilet seat, with a seat cover-up;
(3) FIG. **2** illustrates an embodiment of a perspective view of a bidet washing apparatus installed on an existing toilet seat, with a seat cover down;
(4) FIG. **3** illustrates an embodiment of a perspective view of an exemplary bidet washing apparatus;
(5) FIG. **4** illustrates an embodiment of a perspective view of an exemplary bidet washing apparatus, with dotted lines showing nozzles extended outwards;
(6) FIG. **5** illustrates an embodiment of a fragmentary view of an exemplary bidet washing apparatus illustrating a gate shield protecting the nozzle assembly in a closed position;
(7) FIG. **6** illustrates an embodiment of a fragmentary view of an exemplary bidet washing apparatus illustrating a shield gate protecting the nozzle assembly in an open position;
(8) FIG. **7** illustrates an embodiment of a top plan view of an exemplary embodiment;
(9) FIG. **8** illustrates an embodiment of a bottom plan view of a perspective view of a bidet washing apparatus;

(10) FIG. **9** illustrates an embodiment of a perspective view of an exemplary bidet washing apparatus;
(11) FIG. **10** illustrates an embodiment of a front top view of a bidet washing apparatus with a deodorizer housing unit, deodorizer container and deodorizer actuator;
(12) FIG. **11** illustrates an embodiment of a rear bottom view of a bidet washing apparatus with a deodorizer housing unit, deodorizer container and deodorizer actuator;
(13) FIG. **12** illustrates an embodiment of a front cut-away view of a deodorizing housing unit, deodorizer container and deodorizer actuator;
(14) FIG. **13** illustrates an embodiment of a front exploded view of the housing unit, container, actuator, and a pump
(15) FIG. **14** illustrates an embodiment of a front cut-away view of a bidet washing apparatus with a deodorizer housing unit, deodorizer container and deodorizer actuator;
(16) FIG. **15** illustrates an embodiment of a rear cut-away view of a nozzle assembly, which includes the washing nozzles, the protective shield gate, and the deodorizer spray or drop outlet;
(17) FIG. **16** illustrates an embodiment of a top cut-away view of the nozzle assembly including three inlets, three water tubes, washing nozzles, a deodorizer inlet, deodorizer tube, and a deodorizer spray or drop outlet;
(18) FIG. **17** illustrates an embodiment of an exploded, top cut-away view of the nozzle assembly including three inlets, three water tubes, washing nozzles of the nozzle assembly, deodorizer inlet **126**, deodorizer tube, and a deodorizer spray or drop outlet;
(19) FIG. **18** illustrates an embodiment of a side cut-away view of the deodorizer housing unit, deodorizer container **121**, deodorizer actuator, and a pump when in an inactivated position;
(20) FIG. **19** illustrates an embodiment of a side cut-away view of the deodorizer housing unit, deodorizer container **121**, deodorizer actuator, and a pump when in an activated position;
(21) FIG. **20** illustrates an alternative embodiment of a front top view of a bidet washing apparatus, including a deodorizer container and deodorizer actuator incorporated into the control unit of a bidet washing apparatus;
(22) FIG. **21** illustrates an alternative embodiment of a rear top cut-away view of a bidet washing apparatus, including the deodorizer container and deodorizer actuator incorporated into the control unit of the bidet washing apparatus;
(23) FIG. **22** illustrates another alternate embodiment of a front top view of a bidet washing apparatus, including the deodorizer housing unit, deodorizer container, and the deodorizer actuator incorporated onto the bidet washing apparatus distal to the control unit;
(24) FIG. **23** illustrates another alternate embodiment of a front top view of a bidet washing apparatus, including the deodorizer housing unit, deodorizer container, and the deodorizer actuator incorporated onto the bidet washing apparatus distal to the control unit; and
(25) FIG. **24** illustrates another alternate embodiment of a front top cut-away view of a bidet washing apparatus, including the deodorizer housing unit, deodorizer container, and the deodorizer actuator arranged in the control unit;
(26) FIG. **25**A illustrates an embodiment of a side cut-away view of a deodorizer housing unit, which includes an aerosol container and an actuator configured to deliver a deodorizer solution to a toilet bowl;
(27) FIG. **25**B illustrates an embodiment of a side cut-away view of a deodorizer housing unit, which includes an aerosol container and an actuator configured to deliver a deodorizer solution to a toilet bowl;
(28) FIG. **26** illustrates an embodiment of a side cut-away exploded view of a deodorizer housing unit, which includes an aerosol container and actuator;
(29) FIG. **27** illustrate an embodiment of a side cut-away view of the housing unit, which includes the aerosol container and the actuator, when in an inactivated position; and
(30) FIG. **28** illustrate an embodiment of a side cut-away view of the housing unit, which includes

the aerosol container and the actuator, when in an activated position.

DETAILED DESCRIPTION OF EXEMPLARY EMBODIMENTS

(31) The following description is presented to enable a person of ordinary skill in the art to make and use embodiments described herein. Descriptions of specific devices, techniques. and applications are provided only as examples. Various modifications to the examples described herein will be readily apparent to those of ordinary skill in the art, and the general principles defined herein may be applied to other examples and applications without departing from the spirit and scope of the disclosure. Thus, the disclosure is not intended to be limited to the examples described herein and shown but is to be accorded the scope consistent with the claims.

(32) The word “exemplary” is used herein to mean “serving as an example illustration,” Any aspect or design described herein as “exemplary” is not necessarily to be construed as preferred or advantageous over other aspects or designs.

(33) Reference will now be made in detail to aspects of the subject technology, examples of which are illustrated in the accompanying drawings, wherein reference numerals refer to like elements throughout.

(34) It should be understood that the specific order or hierarchy of steps in the process disclosed herein is an example of exemplary approaches. Based upon design preferences, it is understood that the specific order or hierarchy of steps in the processes can be rearranged while remaining within the scope of the disclosure. Any accompanying method claims present elements of the various steps in, a sample order, and are not meant to be limited to the specific order or hierarchy presented.

(35) The embodiments disclosed herein describe a new, clean and hygienic washing bidet. The various embodiments include one or a plurality of water inlets, a control means housing one or a plurality of control valves to control the flow of water from the water inlets to one or a plurality of water tubes, one or a plurality of washing nozzles, a protective shield gate, and securing unit configured to securing the sanitary washing device to the toilet seat.

(36) The disclosed embodiments directed to clean and hygienic bidet washing apparatus **100** attachable to an existing toilet, for cleaning the body parts of the user sitting on or near the toilet.

(37) As described herein, a “bidet” is a toilet attachment for cleaning the body parts of the user.

(38) As described herein, the term “water inlet” means any structure that may provide water to the bidet washing apparatus.

(39) As described herein, a “control unit” (aka “control panel”) is the housing which has “control switch(s)” thereon controlling the various functionalities of the bidet, including but, not limited to, the flow of water, adjusting the angle of the nozzles, and opening and closing the protective shield gate.

(40) As described herein, “control valves” are, controller parts located inside the control panel housing which control the flow of water or other fluids from the water inlet(s) to one or more “water tubes” by opening, closing, or partially obstructing various passageways.

(41) As described herein, “water tubes” are channels that connect the control valves to a “nozzle assembly,” wherein, the “nozzle assembly” includes a single nozzle or a collection of nozzles including at least one “washing nozzle.”

(42) As described herein, a “nozzle” is a device designed to eject water or other fluids into the surrounding medium as a coherently controlled spray.

(43) As described herein, the “washing nozzle” is the nozzle that can be used to wash the body parts of a user.

(44) As described herein, the “nozzle assembly” may also have other types of nozzles such as a “self-cleaning nozzle,” which is used to clean the nozzle assembly itself, a “toilet cleaning nozzle,” which is used to clean the bidet and/or the toilet, and a “shield cleaning nozzle,” which is used for cleaning the “protective shield gate.”

(45) As described herein, the “protective shield gate” is a structure placed at least partially in front of the nozzle assembly (e.g., between the user and the nozzle assembly) to protect the nozzle

assembly from pollutants.

(46) As described herein, the “protective shield gate” has a “hinged” edge. The term “hinged” here means a joint that allows the turning or pivoting of the gate, by any conventional turning or pivoting mechanism.

(47) As described herein, the term “fluidically coupled” means a connection or a passageway that allows fluid to flow therethrough.

(48) As described herein, the term “reservoir” means a fluid holding tank.

(49) Accordingly, in one embodiment the disclosure provides a bidet washing apparatus attachable to a toilet bowl for cleaning one or more body parts of a user. The apparatus can include one or more water inlets configured to supply water, and a control unit, housing one or more valves fluidically connected to the one or more water inlets, including one or more control switches configured to operate the one or more valves. As such, one or more valves can control water flow from one or more water inlets. The apparatus can further include a nozzle assembly including at least one washing nozzle, fluidically connected to at least one of the one or more valves with one or more water tubes. At least one washing nozzle can be positioned for directing water to one or more body parts of the user. The apparatus can also include a protective shield gate covering at least a portion of at least one washing nozzle, where the protective shield gate is rotatably coupled to the bidet washing apparatus.

(50) According to various embodiments, the protective shield gate can be rotatably coupled to the apparatus along a side or top edge via a hinge, for example, to allow for the manual or electrical opening and closing of the protective shield gate. In this matter, the nozzle(s) are easily accessible for cleaning, removal, replacement, or another adjustment while the protective shield gate is open. In an alternative embodiment, the protective shield gate can be completely removed to similarly provide access to the nozzle(s).

(51) According to another embodiment, the water inlet(s) can be fluidically connected to one or more valves via a single-body connector without any intervening parts or joints, which results in a more robust, long-lasting, bidet washing apparatus, since leaks or other damage to the fluidic couplings are less likely to occur.

(52) Referring to FIG. **1** and FIG. **2**, the bidet washing apparatus **100** of the disclosure can be mounted on a toilet bowl **110** using securing mechanisms **105***a* and **105***b*. Any conventional securing unit can be implemented, e.g., one or more screws. A toilet seat **112** can pivot around and can be connected to a rear portion of the toilet bowl **110**. On the rear portion of the toilet bowl, **110** can be mounted a refillable toilet tank **109**, in which an amount of water can be stored. In certain embodiments, toilet tank **109** can be used as the water source for the bidet washing apparatus **100** by a fluidic connection. On the bidet washing apparatus, **100** can be mounted a nozzle assembly **101**, which includes at least one washing nozzle (not shown) for washing the body parts of the user sitting on or near the toilet bowl **110**. The body of the bidet washing apparatus can be made of any suitable material, including but not limited to, plastics, polymers, reinforced polymeric materials, wood, metal and the like, and any combination thereof.

(53) FIG. **3** shows one exemplary embodiment of a bidet washing apparatus **100** with two washing nozzles **101***a* and **101***b*, respectively. However, in an installation, a lesser or greater number of nozzles can be used. Each washing nozzle can spray a stream of water upwardly and inwardly, according to various embodiments.

(54) As shown in FIG. **2** and FIG. **3**, a control unit **108** can be provided the easy access for the user, and houses control switches **102***a* and **102***b* for providing operational instructions to the bidet washing apparatus **100**. The depicted example shows two switches **102***a* and **102***b*; however, one of ordinary skill in the art would realize that any number of switches can be provided for performing various operations without departing from the scope of the disclosure. Some examples of operational instructions include, but are not limited to, controlling the flow of water from the water inlet, changing the angle of the washing nozzles, and opening and closing the protective shield gate

(described in further detail below). The type of control switches can be selected from a group including knobs, dials, levers, depressible buttons, or any conventional control mechanism, An installation may have all similar control switches where both control switches **102***a* and **102***b* are knobs.

(55) On the other hand, FIG. **9** shows an embodiment of the disclosure where one of the control switches **102***b* is a knob and the other control switch **102***c* is a lever.

(56) Furthermore, as shown in FIG. **1** and FIG. **2**, the nozzle assembly **101** can have a protective shield gate **104** substantially or partially in front of it. The position of the protective shield gate **104** is such that it can act as a shield between the user's body and/or water in the toilet bowl **110** and the nozzle assembly **101** thus protecting the nozzle assembly **101** from pollutants during use.

(57) Certain aspects of the bidet washing apparatus will be detailed hereinafter concerning FIGS. **3-9**.

(58) FIG. **3** shows a front perspective view of one embodiment described herein. Referring to FIG. **3**, the bidet washing apparatus **100** includes the water inlets **103***a* and **103***b* to feed water into the bidet. The water inlet can be controlled by the user using the control switches **102***a* and **102***b* situated on the control unit **108**. The water from the water inlets **103***a* and **103***b* can be ultimately provided to the nozzle assembly **101** via tubes (as described in greater detail concerning FIG. **7**, for example). The nozzle assembly shown in this aspect of the disclosure has two washing nozzles **101***a* and **101***b*. The protective shield gate **104** protects the nozzle assembly **101** from excrement and pollutants as described above. In this example, the protective shield gate **104** is positioned in front of the nozzle assembly **101** of the bidet such that it is between the user sitting on the toilet seat, for example, and the nozzle assembly. Hence, when the user is using, the toilet, the nozzles are shielded behind the protective shield gate **104** and do not become polluted.

(59) Referring to FIGS. **5-6**, the protective shield gate **104** can be movable along a hinged edge **111** to provide for further hygiene. The protective shield gate **104** rotates and thus can be manual, for example, opened (FIG. **6**) and closed (FIG. **5**), after using the toilet to clean any minute leftover pollutants on the outer covering of the nozzle assembly to ensure complete cleanliness. In the embodiment, the user can open and close the gate manually and, thus, the gate can stay in the opened or closed position that the user places the gate. In certain embodiments, the gate can be opened and dosed by an electrical signal using a control switch located on the control unit **108**, which can allow the gate to remain open until the user closes the gate via the control switch, so the user can clean the nozzle. In other embodiments, the hinged edge **111** is on the top of the protective shield gate **104**, and not, on the side edge as shown in the illustrative FIGS. **5-6**. hi. yet other embodiments, the user may be able to completely remove the protective shield gate **104** for cleaning the nozzle(s) and reattach it after cleaning. Of course, one of ordinary skill in the art would understand that the hinged edge could comprise any rotatable joint mechanism that allows for, the rotation of the protective shield gate **104** to provide efficient access to the nozzle(s). If the protective shield gate **104** is completely removable, a grooved and slidable mechanism can be employed so that the protective shield gate **104** can slide in and out to be attached and removed. Of course, other mechanisms can be utilized for removably attaching the protective shield gate **104**, e.g., a magnet or a snap structure.

(60) In certain embodiments, the protective shield gate **104** has a flap portion perpendicular to the protective shield gate **104** such that it covers the bottom of the nozzle assembly **101**. Additionally, the flap can have a spring mechanism such that it is pushed out and aligns with the protective shield gate **104** by the force of the water stream when water flows out of the nozzle assembly **101**. When the water flow stops, the flap can spring hack into its original position perpendicular to the shield gate **101**.

(61) The protective shield gate **104** of the disclosure can be made from a material selected from plastic, metal, a material having anti-microbial properties, and material with increased pollutant repellant properties.

(62) In certain embodiments, the angle of the washing nozzles can be adjusted using a control switch located on the control unit **108**. Thus, when a user wants to clean certain body parts, water can be sprayed on the desired body part by adjusting the angle of the washing nozzle(s). As shown in FIG. **3**, the height of the protective shield gate is such that it allows for an uninterrupted spray of water from the nozzle assembly **101**, since the nozzle assembly **101** can extend beyond the bottom edge of the protective shield gate **104**.
(63) FIG. **4** shows another embodiment wherein the height of the protective shield gate is equal to or greater than that of the washing nozzles **101***a* and **101***b*. Here, the washing nozzles **101***a* and **101***b* are housed within an outer covering including a spring mechanism for pushing the washing nozzles out when water flows through the washing nozzles such that the water flow is not interrupted by the protective shield gate **104**. Each washing nozzle includes an outer covering and an inner nozzle operated slidably hack and firth with hydraulic pressure of the supplied washing water by an instruction from the control unit **108**. During the use of the washing nozzles, the nozzles are extended from their outer covering below the length of the shield gate by the hydraulic force of the washing water, and water is sprayed on the user for cleaning purposes. After use, when the water flow is stopped, the nozzles are retracted in their outer covering which is hidden behind the shield gate. In certain other embodiments, the user may control the movement of the washing nozzle by using the control unit **108**, instead of the hydraulic pressure. When an instruction of a washing operation is given by the control unit **108**, a washing nozzle driving unit is activated to advance the nozzle. The washing nozzle angle can also be adjusted by an instruction given by the control unit **108** to position the nozzle for cleaning. Thus, the washing nozzle can reach the user's desired washing position by the combined advancement of the nozzle and/or the angular positioning.
(64) According to the embodiment, nozzle assembly **101** includes at least one washing nozzle in yet another embodiment, the bidet washing apparatus **100** further includes a self-cleaning nozzle for cleaning the nozzle assembly itself. The self-cleaning nozzle can be positioned to spray water onto the nozzle assembly **101** and/or washing nozzle(s) before and/or after the usage for additional hygiene. The self-cleaning nozzle can be adapted to be controlled by the control unit **108**, and thus provides an additional hygiene level.
(65) Another embodiment includes a toilet cleaning nozzle for cleaning the toilet and the bidet before and after use of the toilet. The toilet cleaning nozzle can be positioned to spray water on the toilet bowl **110** and/or the bidet washing apparatus **100** and can be controlled by the control unit to provide additional hygiene. Yet, another embodiment includes a shield cleaning nozzle for cleaning the protective shield gate **104**. The shield cleaning nozzle can be similarly controlled by the control unit **108**. Additionally, the shield cleaning nozzle can be positioned to clean the protective shield gate **104** in an open and/or closed position.
(66) Any or all of the washing nozzles can be connected to the nozzle assembly **101** via a ball joint, for example. which could allow the user to manually swivel a washing nozzle around 360 degrees, to direct the spray of water in a desired and precise direction. Of course, other types of joints and connectors could be implemented to allow for the manual swivel or direction correction, as desired by the user to spray water to the desired body part, for example.
(67) Moreover, according to an exemplary embodiment, one or more of the washing nozzle(s) **101***a* and **101***b* can be connected to the nozzle assembly **101** by a mechanism allowing for the easy removal of the nozzle(s) **101***a* and **101***b*. For example, the washing nozzle(s) **101***a* and **101***b* can slide into place via a grooved portion of the nozzle assembly **101** or could otherwise snap into place. Any conventional mechanism of removably attaching the nozzle(s) **101***a* and **101***b* can be implemented, so that the user can swap the nozzle(s) **101***a* and **101***b* with other nozzles or increase or reduce the number of washing nozzle(s) **101***a* and **101***b* connected to the nozzle assembly **101**.
(68) An exemplary water supply system to the nozzle assembly **101** will be detailed hereinafter concerning FIGS. **7-8**. The control unit **108** can house the control valves **106***a* and **106***b* (as shown

in FIG. **8**), to control the flow of water to the water tubes and has the control switches **102***a* and **102***b*, for giving instructions to the control valves. Two control valves and control switches are depicted for exemplary purposes, but it should be understood that any number of control valves and corresponding switches can be employed.

(69) The control valves **106***a* and **106***b* can be situated at the entrance to the water tubes **107***a*, **107***b*, and **107***c* in this example. The control valves **106***a* and **106***b* are designed to open, close, or partially obstruct the water inlet **103***a* opening into the water tubes **107***a*, **107***b*, and **107***c*, such that the volume of the water flowing through any tube at any given time can be easily controlled by the user by giving simple instructions through the control switches. The water tubes **107***a*, **107***b*, and **107***c* connect the control valves **106***a* and **106***b* at one end to the nozzle assembly **101** at the other end. Thus, the control valves **106***a* and **106***b* can effectively control the volume of water flowing to the nozzle assembly **101**. In the embodiment, one water tube **107***b* passes through the back of the bidet washing apparatus **100**, and two water tubes **107***a* and **107***c* pass through the front of the bidet washing apparatus **100**. However, it is to be noted that in an embodiment, more than one water tube could pass through the back of the bidet washing apparatus **100**, and the number of water tubes passing through the front of the bidet washing apparatus **100** could be more or less than two.

(70) According to an embodiment, the bidet washing apparatus **100** can include a vacuum breaker (not depicted), which can be situated at various locations within the bidet washing apparatus **100**. The vacuum breaker can be located anywhere between the water supply (e.g., the water tank supplying water to the toilet bowl) and the washing nozzle(s) e.g., **101***a* and **101***b*) output. The vacuum breaker can be intended to halt the flow of water that is not expelled by the washing nozzle(s) back into the water supply. According to one exemplary embodiment, the vacuum breaker(s) can be housed within the control unit **108**, located between a valve **106***a* and **106***b* and the nozzle assembly **101**; however, one of ordinary skill in the art would realize that various locations of one or more vacuum breakers can be implemented within the scope of this disclosure to perform the desired function.

(71) In some embodiments, a bidet washing apparatus is provided that includes a mechanism for delivering an aliquot of a liquid deodorizing solution, a liquid antibacterial and/or a liquid antiseptic solution into a toilet bowl or combinations thereof. For example, a user may dispense a spray, a mist, or one or two drops from the mechanism into the bowl before use. The deodorizing solution forms a thin film on top of the surface of the water that traps and eliminates embarrassing odors before they escape into the air. In addition, as matter enters the toilet bowl water, it is coated with the deodorizing solution thereby eliminating odor.

(72) Common ingredients for a liquid deodorizing solution includes but are not limited to essential oils, alcohols, and water. In other embodiments, the liquid deodorizing solution includes one or more cleaners, for example, alkalis, acids, detergents, abrasives, sanitizers, and spirit solvents.

(73) Useful essential oils include but are not limited to lavender (*Lavandula officinalis*), rosemary (*Rosmarinus officinalis*), tea tree (*melaleuca alternifolia*), lemon (*citrus limonum*), peppermint (*Mentha piperita*), *eucalyptus* (*eucalyptus globulus, eucalyptus sideroxylon* and *eucalyptus* torquate), clove (syzgium *aromaticum*), chamomile (*anthemis nobilis*), frankincense (*boswellia carterii*), myrrh (*commiphora* myrrha), grapefruit (*citrus paradisi*), oregano (*Origanum vulgare*), and ginger (*Zingiber officinale*), *hamamelis virginiana* extract, *rosa* damascene flower oil, *Lavandula angustifolia* oil, aloe *barbadensis* leaf extract, *mentha arvensis* leaf oil, *Sophora japonica* flower extract, and combinations thereof.

(74) Useful alcohols include but are not limited to C1-C6 alcohols, for example, methanol, ethanol, propanol, isopropanol, and the like) and combinations thereof. Other ingredients include but are not limited to plant extracts, antimicrobial organic salts, essential oil solubilizer, emulsifier, surfactant, and water and combinations thereof.

(75) FIG. **10** illustrates an embodiment of a front top view of a bidet washing apparatus **100**, which includes a deodorizer housing unit **120**, a container **121**, and a deodorizer actuator **122**. Also shown

in this figure is the control unit **108**, control switches **102**_a_ and **102**_b_, the nozzle assembly **101**, washing nozzles **101**_a_ and **101**_b_. Located in proximity to the washing nozzles is the deodorizer spray or drop outlet **123**, which can transfer a liquid deodorizer solution from the deodorizer container into a toilet bowl. In this embodiment, the protective shield gate has been omitted for clarity.

(76) In embodiments, the container **121** can be reversibly or permanently attached to the housing unit **120**, for example, by being snapped on, screwed on, or bolted on; or can be permanently attached to the housing unit by being molded or glued thereto or by using any other means known in the art. The housing unit **120** can also be reversibly or permanently attached to the control unit **108** of the bidet washing apparatus **100**, for example, by being snapped on, screwed on, bolted on; or permanently attached to the control unit by being molded or glued thereto or by using any other means known in the art.

(77) In other embodiments, the housing unit **120**, container **121**, and deodorizer actuator can be directly incorporated into the control unit **108** of the bidet washing apparatus **100** as shown below. For both configurations, when the actuator is engaged, e.g., by manually pushing the actuator down, a pump is actuated and an aliquot of the liquid deodorizer solution present in the deodorizer container can be delivered by spray, mist, or drop into a toilet bowl.

(78) As used herein, an “actuator” includes but is not limited to a push-button, twist-button, pull up and push down button, a slide button, a plunger button, a lever, a knob, or other similar devices.

(79) FIG. **11** illustrates an embodiment of a rear bottom view of a bidet washing apparatus **100**, which includes a housing unit **120**, a deodorizer container **121**, and an actuator **122** attached thereto. Also shown is the control unit **108**, control switches **102**_a_ and **102**_b_, water inlets **103**_a_ and **103**_b_, the nozzle assembly **101** including wash nozzles **101**_a_ and **101**_b_, the protective shield gate **104**, and the deodorizer spray or drop outlet **123**.

(80) FIG. **12** illustrates an embodiment of a front cut-away view of the housing unit **120** (without the control unit shown for simplicity). As shown, the bidet housing unit includes the container **121** and the actuator **122** attached thereto.

(81) FIG. **13** illustrates an embodiment of a front exploded view of the housing unit **120**, the container **121**, the actuator **122**, and a pump **130** including a pump chamber **131**, a pump inlet tube **132**, and a pump outlet tube **137** and pump outlet tube opening **138**, and control unit **108**.

(82) FIG. **14** illustrates an embodiment of a front cut-away view of a bidet washing apparatus **100**, which includes a housing unit **120**, a container **121**, and an actuator **122** attached thereto. Also shown is the protective shield gate **104**, and the deodorizer spray or drop outlet **123**. As shown, a deodorizer tube **123 124** runs from the housing unit **120** to the nozzle assembly **101**. The deodorizer tube **123 124** can run across the bidet washing apparatus **100** along a similar path as water tubes **107**_a_, **107**_b_, and **107**_c_ to the nozzle assembly **101** and into the deodorizer spray or drop outlet **123**. In embodiments, when the actuator is engaged, e.g., by manually pushing the actuator down, a pump in the deodorizer housing unit **101** is actuated and an aliquot of the liquid deodorizer solution present in the container **121** can be delivered through the pump outlet **137** in the housing unit **120** through the deodorizer tube **124** and to the deodorizer spray or drop outlet **123** for delivery into a toilet bowl. In some embodiments, the pump is a bottle pump. In other embodiments, an aerosol spray device can be configured to deliver a deodorizer solution.

(83) FIG. **15** illustrates an embodiment of a rear cut-away view of a nozzle assembly **101**, which includes the washing nozzles **101**_a_ and **101**_b_, the protective shield gate **104**, and the deodorizer spray or drop outlet **123**. In embodiments, the deodorizer spray or drop outlet can be located behind the protective shield gate. In other embodiments, the deodorizer spray or drop outlet can be incorporated above or in front of the protective shield gate.

(84) FIG. **16** illustrates an embodiment of a top cut-away view of the nozzle assembly **101**, which includes inlets **125**_a_, **125**_b_, and **125**_c_ for connecting water tubes to the washing nozzles; and deodorizer inlet **126** for connecting deodorizer tube to the deodorizer spray or drop outlet.

(85) FIG. **17** illustrates an embodiment of an exploded, top cut-away view of the nozzle assembly **101**, which includes inlets **125***a*, **125***b*, and **125***c* for connecting water tubes to the washing nozzles of the nozzle assembly; and deodorizer inlet **126** for connecting the deodorizer tube to the deodorizer spray or drop outlet **123**.
(86) FIGS. **18** and **19** illustrate an embodiment of a side cut-away view of the housing unit **120**, which includes the container **121**, the actuator **122**, and a pump **130** when in an inactivated and an activated position, respectively.
(87) As shown in these figures, the pump **130** can be located within the housing unit **120** near or above the container **121**. The pump **130** includes a pump chamber **131**, a pump inlet tube **132**, a ball **133**, a spring **134**, a piston **135**, a pump barrel **136**, and a pump outlet tube **137**.
(88) As shown in FIGS. **18** and **19**, when the actuator **122** is engaged, e.g., by manually pushing the actuator down, the piston **135** moves down and compresses the spring **134** in the pump barrel **136**. The upward air pressure draws the ball **133** upwards along with liquid deodorizer solution in the pump inlet tube **132** and in the container **121** into the pump chamber **131**. Releasing the actuator **122** causes the spring **134** to expand returning the piston **135** and actuator **122** to their initial up position and the ball **133** is returned to its resting position. As a result, the ball **133** covers and closes the pump outlet tube opening **138** and the pump inlet tube **132** preventing the liquid deodorizer solution present in the pump chamber **131** from returning to the container **121**. Pressing down on the actuator **122** again causes another round of drawing the liquid deodorizer solution in the pump inlet tube **132** and in the container **121** into the pump chamber **131**, which causes the liquid deodorizer solution already in the pump chamber **131** to be forced out through the pump outlet tube **137**, into the deodorizer tube **124**, and eventually into the deodorizer spray or drop outlet (not shown) and the toilet bowl.
(89) FIG. **20** illustrates an alternative embodiment of a front top view of a bidet washing apparatus **100**, in which the container **121** and actuator **122** are incorporated into the control unit **108** of the bidet washing apparatus. As shown, the apparatus also includes the protective shield gate **104**, which covers the nozzle assembly **101** and washing nozzles **101***a* and **101***b*.
(90) FIG. **21** illustrates an alternative embodiment of a rear top cut-away view of a bidet washing apparatus **100**, in which the container **121** and actuator **122** are incorporated into the control unit **108** of the bidet washing apparatus.
(91) FIG. **22** illustrates another alternate embodiment of a front top view of a bidet washing apparatus **100**, in which the housing unit **120**, container **121** and the actuator **122** are incorporated onto the apparatus distal to the control unit **108**.
(92) FIG. **23** illustrates another alternate embodiment of a front top view of a bidet washing apparatus **100**, in which the housing unit **120**, container **121** and the actuator **122** are incorporated onto the apparatus distal to the control unit **108**. As shown in this figure, the deodorizer tube **124** extends from the housing unit **120** to the deodorizer spray or droplet outlet **123**, which can extend into a toilet bowl for delivery of the liquid deodorizing solution.
(93) FIG. **24** illustrates another alternate embodiment of a front top cut-away view of a bidet washing apparatus **100**, in which the housing unit **120**, container **121** and the actuator **122** are shown as arranged in the control unit **108**. As shown in this figure, the deodorizer tube **124** extends from the housing unit **120** to the deodorizer spray or droplet outlet **123**, which can extend into a toilet bowl for delivery of the liquid deodorizing solution.
(94) FIGS. **25**A and **25**B illustrate an embodiment of a side cut-away view of a deodorizer housing unit **120**, which includes an aerosol container **140** and an actuator **122** configured to deliver a deodorizer solution to a toilet bowl. Also shown in these figures is the outlet tube **137** and opening **138**, a valve stem **141** on top of the container, gasket **142**, valve cup **143**, spring **144**, valve housing unit **145**, and a dip tube **146**.
(95) In some embodiments, the aerosol container includes a liquified gas, such as butane or propane, which is used as a propellant for delivery of the liquid deodorizer solution. In a typical

aerosol configuration, some of the propellant exists as a gas under pressure above the liquid solution. When the actuator **122** is pressed downwards, this forces the valve cup **143** down and compresses the spring **144** in the valve housing unit **145**. Then, the gas present in the container pushes on the liquid deodorizer solution, forcing the liquid up through the dip tube **146**, past the spring **144** and valve cup **143** and out the valve stem **141** and into the outlet tube **137**. The expelled liquid is a mixture of the liquid deodorizer solution and the liquified gas. Conversely, when the actuator **122** is released, the spring **144** elongates and forces the valve cup **143** up thereby covering the valve stem **141** and preventing escape of the liquid.
(96) FIG. **26** illustrates an embodiment of a side cut-away exploded view of a deodorizer housing unit **120**, which includes the aerosol container **140** and actuator **122**.
(97) In some embodiments, the deodorizer container can be reversibly attachable to the housing unit through a screw-on, threaded type mechanism **168**, or can be snapped on, or can be secured permanently or reversibly or by any other means known in the art. In addition, a gasket can be used between the actuator and the valve stem, which prevents leakage of the deodorizer solution.
(98) FIGS. **27** and **28** illustrate an embodiment of a side cut-away view of the housing unit **120**, which includes the aerosol container **140** and the actuator **122**, when in an inactivated and an activated position, respectively. As shown in these figures, when the actuator **122** is not engaged, the deodorizer solution stays in the aerosol container **140** due to the valve cup **143** closing off the valve stem **141**. By contrast, when the actuator **122** is engaged, e.g., by manually pushing the actuator downwards, the deodorizer solution is expelled from the aerosol container **140** through the dip tube **146**, into the valve housing unit **145**, past the valve cup **143**, into an opening in the valve stem **141**, out through the outlet tube **137**, and into a deodorizer tube (not shown) for delivery to a toilet bowl through a deodorizer spray or drop outlet (not shown) as described above.
(99) While the inventive natures have been particularly shown and described concerning preferred embodiments thereof, it will be understood by those in the art that the foregoing and other changes can be made therein without departing from the spirit and the scope of the disclosure. Likewise, the various diagrams may depict an example architectural or other configuration for the disclosure, which is done to aid in understanding the features and functionality that can be included in the disclosure. The disclosure is not restricted to the illustrated example architectures or configurations but can be implemented using a variety of alternative architectures and configurations. Additionally, although the disclosure is described above in terms of various exemplary embodiments and implementations, it should be understood that the various features and functionality described in one or more of the individual embodiments are not limited in their applicability to the particular embodiment with which they are described. They instead can be applied alone or in some combination, to one or more of the other embodiments of the disclosure, whether or not such embodiments are described, and whether or not such features are presented as being a part of a described embodiment. Thus, the breadth and scope of the disclosure should not be limited by any of the above-described exemplary embodiments.

Claims

1. A bidet washing apparatus, comprising: a control unit fluidically connected to one or more water inlets configured to supply water to the control unit, the control unit including one or more control unit switches configured to operate one or more control unit valves for controlling water flow from the one or more water inlets and/or from one or more control unit outlets; a deodorizer housing unit, wherein the housing unit includes a deodorizer container, and a deodorizer pump; a deodorizer tube fluidically connected to the deodorizer pump; a deodorizer spray or drop outlet fluidically connected to the deodorizer tube; and a nozzle assembly including at least one washing nozzle fluidically connected to at least one of the one or more control unit outlets with one or more control unit outlet to nozzle assembly water tubes.

2. The bidet washing apparatus of claim 1, wherein the deodorizer housing unit, the deodorizer container, and the deodorizer pump are attachable to the control unit.
3. The bidet washing apparatus of claim 1, wherein the deodorizer pump includes a deodorizer actuator.
4. The bidet washing apparatus of claim 1, wherein the deodorizer pump is a bottle pump.
5. The bidet washing apparatus of claim 1, wherein the deodorizer pump includes a pump chamber, a spring, a ball, a piston, a pump tube inlet, and a pump tube outlet.
6. The bidet washing apparatus of claim 1, wherein the deodorizer spray or drop outlet is reversibly attachable to the nozzle assembly.
7. A bidet washing apparatus, comprising: a control unit fluidically connected to one or more water inlets configured to supply water to the control unit, the control unit including one or more control unit switches configured to operate one or more control unit valves for controlling water flow from the one or more water inlets and/or from one or more control unit outlets; a deodorizer container and a deodorizer pump, wherein the deodorizer container and the deodorizer pump are located within the control unit; a deodorizer tube fluidically connected te to the deodorizer pump; a deodorizer spray or drop outlet fluidically connected to the deodorizer tube; and a nozzle assembly including at least one washing nozzle fluidically connected to at least one of the one or more control unit outlets with one or more control unit outlet to nozzle assembly water tubes.
8. The bidet washing apparatus of claim 7, wherein the deodorizer pump includes a deodorizer actuator.
9. The bidet washing apparatus of claim 7, wherein the deodorizer pump is a bottle pump.
10. The bidet washing apparatus of claim 7, wherein the deodorizer pump includes a pump chamber, a spring, a ball, a piston, a pump tube inlet, and a pump tube outlet.
11. The bidet washing apparatus of claim 7, wherein the deodorizer spray or drop outlet is reversibly attachable to the nozzle assembly.
12. A bidet washing apparatus, comprising: a control unit fluidically connected to one or more water inlets configured to supply water to the control unit, the control unit including one or more control unit switches configured to operate one or more control unit valves for controlling water flow from the one or more water inlets and/or from one or more control unit outlets; a deodorizer housing unit, wherein the housing unit includes a deodorizer container, and a deodorizer pump, and wherein the deodorizer housing unit is located distal to the control unit; a deodorizer tube fluidically connected to the deodorizer pump; a deodorizer spray or drop outlet fluidically connected to the deodorizer tube; and a nozzle assembly including at least one washing nozzle fluidically connected to at least one of the one or more control unit outlets with one or more control unit outlet to nozzle assembly water tubes.
13. The bidet washing apparatus of claim 12, wherein the deodorizer pump includes a deodorizer actuator.
14. The bidet washing apparatus of claim 12, wherein the deodorizer pump is a bottle pump.
15. The bidet washing apparatus of claim 12, wherein the deodorizer pump includes a pump chamber, a spring, a ball, a piston, a pump tube inlet, and a pump tube outlet.
16. The bidet washing apparatus of claim 12, wherein the deodorizer spray or drop outlet is reversibly attachable to the nozzle assembly.
17. A bidet washing apparatus, comprising: a control unit fluidically connected to one or more water inlets configured to supply water to the control unit, the control unit including one or more control unit switches configured to operate one or more control unit valves for controlling water flow from the one or more water inlets and/or from one or more control unit outlets; a deodorizer housing unit, wherein the housing unit includes a deodorizer aerosol container; a deodorizer tube fluidically connected to the deodorizer aerosol container; a deodorizer spray or drop outlet fluidically connected to the deodorizer tube; and a nozzle assembly including at least one washing nozzle fluidically connected to at least one of the one or more control unit outlets with one or more

control unit outlet to nozzle assembly water tubes.
18. The bidet washing apparatus of claim 17, wherein the deodorizer housing unit and the deodorizer aerosol container are attachable to the control unit.
19. The bidet washing apparatus of claim 17, wherein the deodorizer spray or drop outlet is reversibly attachable to the nozzle assembly.
20. The bidet washing apparatus of claim 17, wherein the deodorizer housing unit and deodorizer aerosol container are located within the control unit, or are located distal to the control unit.